A Multi-Modal Framework for Quantifying Narrator Reliability in ʿIlm ar-Rijal

Using Semantic Sentiment Analysis and Graph Neural Networks



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Introduction

Hadith authentication is fundamental to Islamic scholarship. Traditional *ʿIlm ar-Rijal* methodology is labour-intensive and increasingly inadequate for the digital era. Over 500,000 narrators have been tracked historically through manual processes.

There is a critical need for an automated, scalable solution that preserves classical methodology while leveraging modern computational approaches. Current manual cross-referencing of vast biographical texts cannot keep pace with digital Hadith scholarship demands.

Problem Background

Hadith comprises two components: *Sanad* (chain of narrators) and *Matn* (textual content). Authentication requires assessing each narrator's moral integrity (*ʿadāla)* and memory precision (*ḍabṭ*) through manual cross-referencing of vast biographical texts such as Taqrīb al-Tahḏīb of Ibn Ḥajar al-ʿAsqalāni and the Tahḏīb al-Kamāl of al-Mizzī.

Key challenges include Arabic orthographic ambiguity, name variations across sources, and temporal/geographical verification complexities.

Critical gap: **No end-to-end automated reliability scoring system exists that preserves classical methodology while enabling digital-scale analysis.**



Problem Statement

Current NER models successfully extract narrator names from Arabic texts, but they have not yet quantified reliability based on *Jarḥ wa Taʿdīl* terminology. Graph-based approaches visualise narrator networks but lack trust scores or semantic understanding of classical evaluation terms.

No integrated framework currently exists that combines semantic sentiment analysis of classical terms, structural graph analysis, and spatiotemporal validation into a unified reliability scoring system for *ʿIlm ar-Rijal* scholarship.

Existing Solutions & Limitations

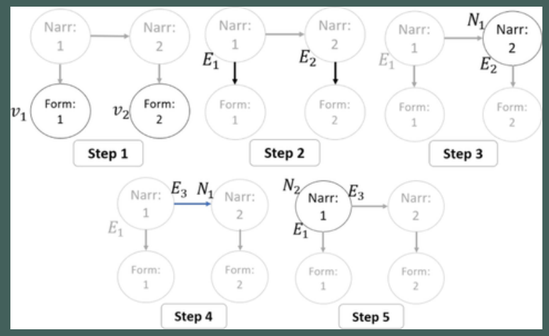
Arabic NER Advances

AraBERT and MARBERT achieve high-accuracy Arabic Named Entity Recognition for narrator extraction (Mahmoud et al. 2022). However, these models **identify names without quantifying reliability based on classical *Jarḥ wa Taʿdīl* terminology.**



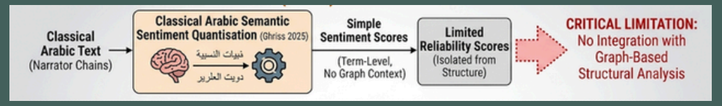
Knowledge Graph Approaches

NarratorsKG constructs narrator network graphs (Mahmoud et al. 2024). Oumer (2025) enables Hadith-to-graph conversion. Mosa (2025) proposes KG-Transformer framework. Limitation: **None provide end-to-end reliability grading.**



Semantic Analysis Gap

Ghriss (2025) introduces semantic sentiment quantisation for classical Arabic terms. Critical limitation: No integration with graph-based structural analysis. **No existing framework combines NER, KG, GNN, and semantic sentiment for complete narrator reliability scoring.**



Proposed Solution

End-to-End Pipeline

First complete automated framework: **extraction → disambiguation → scoring → grading**. No existing solution provides this full pipeline from raw biographical texts to final Hadith reliability classification.

Multi-Modal Integration

Combines NER + Knowledge Graph + Graph Neural Networks + Semantic Sentiment Analysis. Probabilistic trust scores (0.0-1.0) per narrator with edge modifiers for transmission risks including ambiguous attribution (*tadlīs*), memory lapses, and geographical impossibilities.

Explainable & Validated

Evaluation against *Kutub as-Sitta* (the Six Canonical Hadith Collections) ground-truth collections. Explainable AI outputs directly tied to classical *Jarḥ wa Taʿdīl* terminology, ensuring transparency and scholarly acceptance.

Research Objectives

Objective 1: NLP Classification

To develop an NLP classification system that maps classical *Jarḥ wa Taʿdīl* terminology to a quantitative reliability index, enabling systematic conversion of scholarly assessments into numerical trust scores (0.0-1.0).

Objective 2: Knowledge Graph & GNN

To build a comprehensive Knowledge Graph encoding teacher-student transmission links with narrator attributes, apply Graph Neural Network analysis for structural pattern detection, and implement geospatial validation of narrator travel claims.

Objective 3: Framework Evaluation

To evaluate the complete framework against canonical Hadith gradings from Kutub as-Sitta, measuring accuracy across four classification categories: *Ṣaḥīḥ, Ḥasan, Ḍaʿīf, and Mawḍūʿ.*

Problem Scope

Data Sources

Taqrīb at-Tahdhīb

Tahdhīb al-Kamāl

AR-Sanad 280k

Kutub as-Sitta

Ṣaḥīḥ al-Bukhārī

Ṣaḥīḥ Muslim

Sunan Abī Dāwūd

Jāmiʿ at-Tirmiḏī

Sunan an-Nasāʾī

Sunan Ibn Mājah

Technical Scope

Arabic NER, entity disambiguation, reliability scoring, geospatial mapping.

Targeted Scale

- ~1,500 narrators
- ~1,000 *isnād* chains

Out of Scope

- *Matn* criticism (content analysis of Hadith text)
- *Shīʿa* methodological traditions
- Real-time production deployment

Research Gaps from Literature Review

Findings

Key foundations spanning Hadith authentication methodology, Arabic natural language processing, knowledge graph construction, graph neural networks, and semantic sentiment analysis for narrator reliability assessment.

Justifications

Current work is limited to extraction or graph construction only. No full reliability scoring pipeline has been published. NLP and graph methodologies remain separate disciplines. No standardised evaluation exists against canonical Hadith collections.

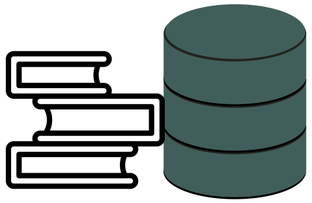
SOTA Gap

Oumer (2025) excludes narrator reliability scoring entirely. This project addresses the void by adding a quantitative trust scoring layer, integrating NER, Knowledge Graphs, GNNs, and semantic sentiment into one end-to-end framework.

Previous Similar Researches

Author(s) & Year	Topic / Focus Area	Methodology / Tools	Key Contributions & Limitations
Mosa (2025)	Narrator Disambiguation / KG-Transformer	Pretrained Language Models (PLMs), Knowledge Graph-Transformer	Synergized structure and semantics for narrator node embeddings; achieved ~97% accuracy on AR-Sanad 280K-v2. Did not assign reliability scores.
Muther et al. (2023)	Named Entity Recognition (NER)	Transformer-based Arabic NER	Precise extraction and disambiguation of narrator identities from classical texts.
Mahmoud et al. (2024)	Structural Integrity & Networks	Graph Neural Networks (GNN), NarratorsKG, Graph Query Embeddings	Introduced narrator identification utilizing graph query embeddings and a knowledge graph. Did not incorporate semantic richness of biographical texts.
Ghriss (2025)	Reliability Scoring / Sentiment Quantisation	Semantic Sentiment Quantisation, Iterative Co-Update Algorithm (ICUA)	Translated qualitative scholarly evaluations into a continuous reliability index (0.0 to 1.0) and computed chain-level scores.
Mahmoud et al. (2022)	Narrator Disambiguation	AraBERT, AR-Sanad 280K dataset	Established a new baseline for narrator disambiguation using large-scale datasets and deep bidirectional representations.
Oumer (2025)	Graph Representation / Isnād Modelling	AraT5, CAMELBERT-CA	Turned Hadith text into relational tuples and graph representations. Explicitly left out narrator reliability scoring and entity linking.
Alghamdi et al. (2025)	Entity Extraction & Fraud Detection	Transformer-based models (AraBERT, MARBERT)	Demonstrated PLMs significantly outperform previous iterations in identifying fraudulent content and extracting technical entities. Did not feed outputs into KG/GNN.
Mghari et al. (2024)	Semantic Sentiment Quantisation	Narrator2Vec, Word Embeddings	Used word embeddings to create dense representations of a narrator's standing, mapping technical terms to reliability weights.
Ahmad et al. (2025)	Isnād Analysis & Reliability	Machine Learning	Confirmed technical viability of automating isnād analysis but noted a critical lack of frameworks capable of quantifying narrator reliability.
Farooqi et al. (2024)	Isnād Modelling	Structured transmission datasets	Contributed foundational work on structured transmission datasets, but highlighted lack of shared benchmark protocols for end-to-end evaluation.
Najeeb (2022)	Isnād Modelling	Sequence-based isnād modelling	Contributed to resolving identities across centuries, though focused primarily on sequence-based extraction.

Phase A:
Data Ingestion



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01

Data Sources

Primary biographical texts including Taqrīb at-Tahdhīb and Tahdhīb al-Kamāl. AR-Sanad 280K dataset for training data. Kutub as-Sitta canonical collections for ground-truth validation.

02

Preprocessing

UTF-8 normalization and Unicode NFKC standardization. Diacritic stripping for consistency. Farasa tokenization for Arabic text. *Isnād-matn* segmentation to separate chain from content.

03

Output

Cleaned, tokenized corpus containing approximately 1,500 narrators and 1,000 *isnād* chains. Structured JSON format for downstream NER and Knowledge Graph construction phases.

Phase B:

NER & Knowledge Graph Construction

Fine-tune AraBERT v2 on AR-Sanad 280K dataset using BIO tagging schema for precise narrator extraction. Build NetworkX Knowledge Graph with comprehensive node attributes including name, *nasab*, *kunya*, *nisba*, dates, *tabaqa* (generation), travels, and *Jarḥ wa Taʿdīl* assessments.

Teacher-student relationships form directed edges in the graph structure. Entity disambiguation combines Levenshtein distance matching with graph neighborhood analysis to resolve Arabic name variations and orthographic ambiguities across biographical sources.



NER Extraction Pipeline

Implement Tabaqa generational classification system spanning 12 tiers from 622-870 CE. Each narrator node receives temporal positioning enabling chronological validation of transmission chains and detection of anachronistic links.

Knowledge Graph Architecture

Graph construction integrates multiple biographical sources including Taqrīb al-Tahdhīb and Tahdhīb al-Kamāl. Node attributes enable multi-dimensional analysis combining textual, relational, and temporal features for comprehensive narrator profiling.

Phase C:
Probabilistic Trust Scoring

Semantic sentiment mapping converts classical *Jarḥ wa Taʿdīl* terminology into quantitative reliability weights on a 0.0-1.0 scale. This preserves scholarly nuance while enabling computational analysis.

Chain scoring calculates the product of individual node scores multiplied by edge risk modifiers, accounting for *tadlis* (concealment), memory degradation, and geographical impossibilities. A 2-layer Graph Convolutional Network validates structural and temporal integrity across the *isnād* network.

Term (Arabic)	Transliteration	Grade	Base Weight
أضبط الناس	*Aḍbaṭ an-nās*	Select highest *taʿdīl*	1.00
ثقة ثقة	*Thiqa thiqa*	Highest *taʿdīl*	0.95
ثقة ضابط	*Thiqa ḍābiṭ*	Trustworthy and precise	0.90
ثقة	*Thiqa*	Strong *taʿdīl*	0.85
صدوق	*Ṣadūq*	Moderate *taʿdīl*	0.70
ليّن	*Layyin*	Mild *jarḥ*	0.20
كذّاب	*Kaḏḏāb*	Severe *jarḥ*	0.00

GNN & Geospatial Validation

The GNN architecture propagates reliability information through the narrator network, identifying anomalous chains and reinforcing trust scores for well-connected, historically verified transmission paths.

Geospatial validation using Geopy and Folium verifies whether narrator travel patterns and meeting locations are historically plausible. This adds a critical verification layer beyond textual analysis alone.

Phase C:
Evaluation Protocol

Metric	Purpose
Accuracy	Overall proportion of correctly classified Hadiths.
Precision/Recall/F1	Per-class performance for each grading category.
RMSE	Root Mean Squared Error between predicted chain scores and expected ordinal grades.
Cohen’s Kappa	Inter-rater agreement between automated and classical human grades.

01

Ground Truth Dataset

Pre-graded Hadith from Kutub as-Sitta (Sahih Bukhari, Sahih Muslim, Sunan Abu Dawud, Jami at-Tirmidhi, Sunan an-Nasa'i, Sunan Ibn Majah) serve as canonical reference for validation.

02

Performance Metrics

Accuracy, Precision, Recall, F1-score for classification performance. RMSE for continuous reliability scores. Cohen's Kappa for inter-rater agreement between framework grades and canonical classifications.

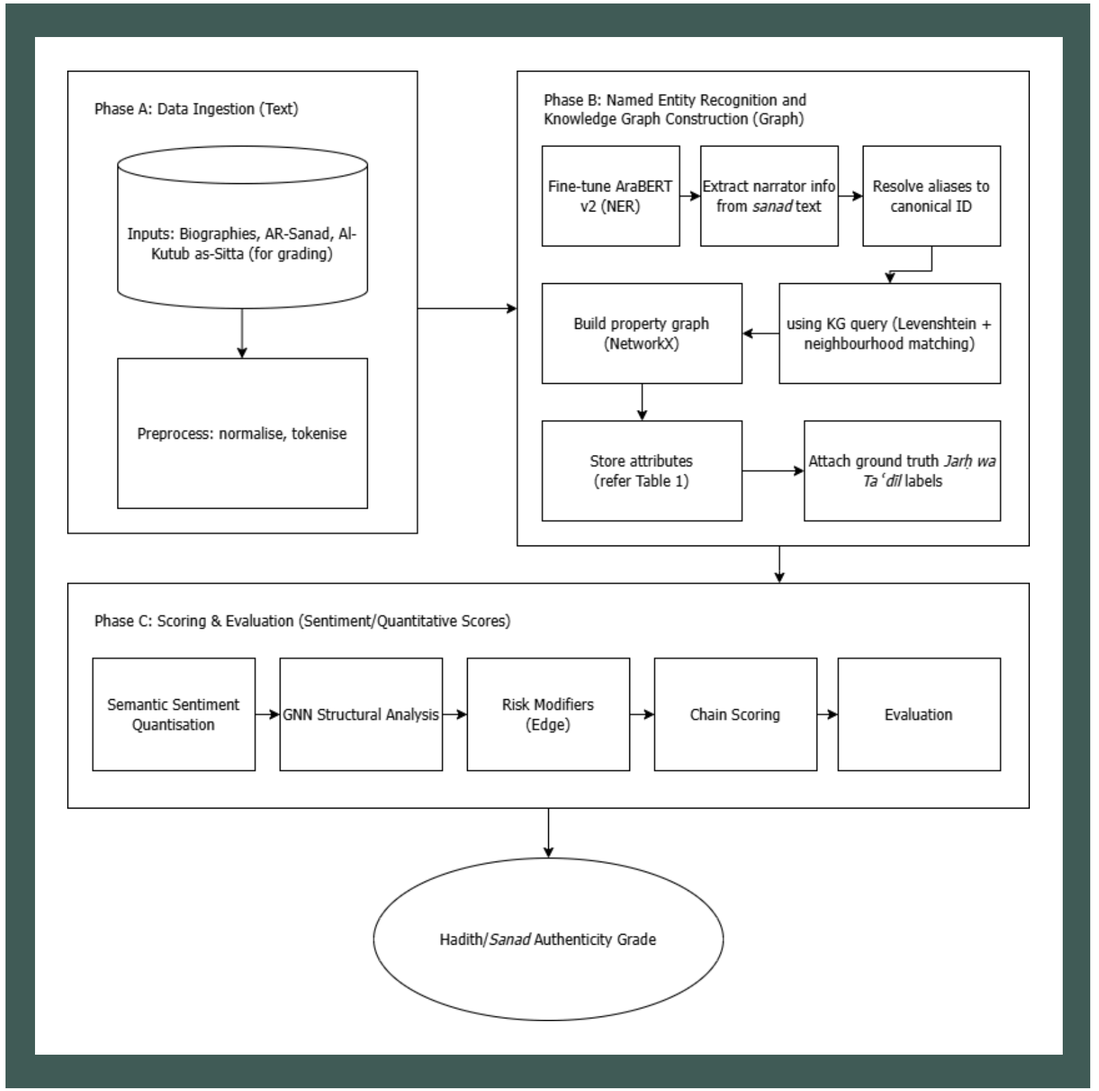
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Baselines & Targets

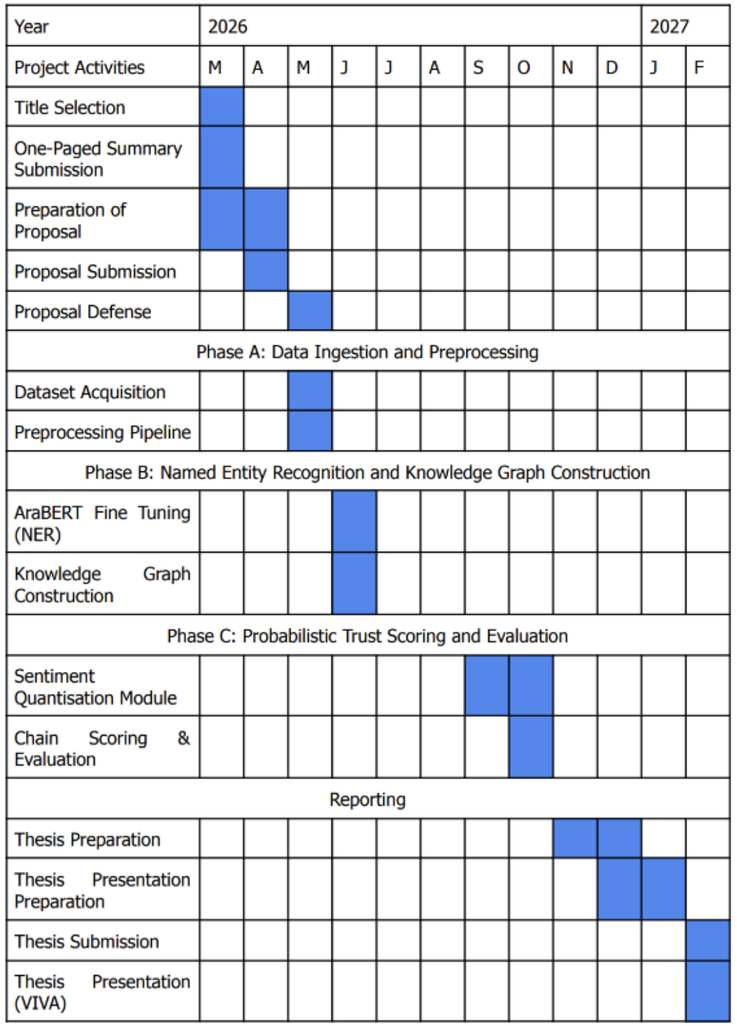
Baselines: Naive classifier, lexical lookup, full pipeline comparison. Performance targets: F1 ≥ 0.90 for NER extraction tasks, Cohen's Kappa ≥ 0.75 for Hadith grading agreement.

Flowchart of Activities

Visual representation of the complete end-to-end pipeline from raw data ingestion through final Hadith grading output.



Week	Milestone	Deliverable
2-2025/2026		
1-2	Project Initialisation	Finalised project summary; supervisor approval
3-4	Literature Review	Completed Chapter 2; dataset acquisition
5-6	Methodology Design	Completed Chapter 3; proposal draft
7	Proposal Defence	Oral defence and proposal submission
8-9	Phase A Implementation	Cleaned corpus; preprocessed JSON dataset
10-11	Phase B Implementation	Trained NER model; populated Knowledge Graph
1-2026/2027		
1-2	Phase C Implementation	Trust scoring module; preliminary evaluation results
3-4	Thesis Report Preparation	Written report drafting
5-6	Thesis Presentation Preparation	Final thesis presentation drafting



Expected Contributions

End-to-End Framework

First end-to-end automated Hadith grading framework that processes biographical data through extraction, disambiguation, scoring, and final grading, a complete pipeline unprecedented in the field.

Interdisiplinary Value

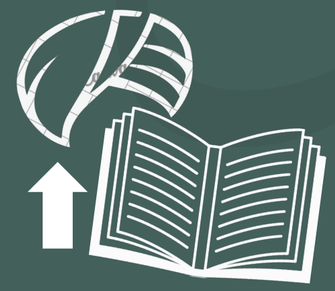
Demonstrates how multi-modal data science integration (NLP and GNN) can be applied to classical knowledge verification systems, mitigating epistemological risks in the digital landscape.

Scholarly Impact

Decision-support tool designed to assist scholars, not replace them. Reproducible, open methodology with standardized evaluation protocol enabling future research validation and extension.







Conclusion

- Clear problem with real-world impact addressing digital Hadith authentication needs.
- Strong understanding of existing solutions and their limitations.
- Novel end-to-end reliability scoring pipeline integrating NER, KG, GNN, and semantic sentiment analysis.

- Research depth demonstrated through tuning, optimisation, and GNN analysis.
- Objectives clearly mapped to methodology phases.
- Framework is explainable, reproducible, and epistemologically sound.

Project Importance

- **Addressing the Scalability Gap**: Modern digital dissemination of Hadith has outpaced manual authentication methods. This framework addresses the "compounding challenge" of expert-dependent labour in 'Ilm ar-Rijal.
- **Beyond Graph Construction**: Fills a critical research gap by moving from simple extraction to end-to-end quantified reliability scoring, a feature currently missing in the SOTA.

Thank You

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